

[illegible]

U1 ES1 ES2-35ROM-1U-N16R2

35

EN

1 IN

2 IOD

3 IOD

4 IOD

5 IOD

6 LCB_IN

7 LCB_IN

8 LCB_IN

9 LCB_IN

10 LCB_IN

11 LCB_IN

12 LCB_IN

13 LCB_IN

14 LCB_IN

15 LCB_IN

16 LCB_IN

17 RXD

18 RXD

19 RXD

20 RXD

21 RXD

22 RXD

23 RXD

24 RXD

25 TXD

26 TXD

27 TXD

28 TXD

29 TXD

30 TXD

31 TXD

32 TXD

33 TXD

34 TXD

35 TXD

GND

The diagram shows the pinout for a 32-pin connector. The pins are numbered 1 to 32. The connections are as follows:

- Pin 1:** +5V
- Pin 2:** J5 DISPLAY
- Pin 3:** J5 DISPLAY
- Pin 4:** J5 DISPLAY
- Pin 5:** J5 DISPLAY
- Pin 6:** J5 DISPLAY
- Pin 7:** J5 DISPLAY
- Pin 8:** J5 DISPLAY
- Pin 9:** J5 DISPLAY
- Pin 10:** J5 DISPLAY
- Pin 11:** J5 DISPLAY
- Pin 12:** J5 DISPLAY
- Pin 13:** J5 DISPLAY
- Pin 14:** J5 DISPLAY
- Pin 15:** J6 NAV_SW
- Pin 16:** J6 NAV_SW
- Pin 17:** J6 NAV_SW
- Pin 18:** J6 NAV_SW
- Pin 19:** +3V3
- Pin 20:** GND
- Pin 21:** J7
- Pin 22:** J7
- Pin 23:** J7
- Pin 24:** J7
- Pin 25:** J7
- Pin 26:** J7
- Pin 27:** J7
- Pin 28:** J7
- Pin 29:** J7
- Pin 30:** J7
- Pin 31:** J7
- Pin 32:** J7

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The schematic diagram illustrates the LED driver circuit for the LED strip. It is powered by a 5V USB supply. The circuit consists of two MOSFETs, SSR1 and SSR2, which drive two channels of the LED strip. Each channel includes a 10k pull-up resistor (R6, R10), a 100k gate resistor (R7, R11), and a 100ohm drain resistor (R8, R12). The MOSFETs are controlled by the USB's D+ and D- lines through a 1k resistor (R9). The LED strip is connected to the MOSFET drains and ground. The circuit is labeled with component values and pin numbers.

WDT_KICKgerbator One-shot, every rising edge on WDT_KICK restarts a ~2.2 s window, so Q stays high only while firmware keeps kicking. Q holds Q3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies SSR_EN – the +5V rail feeding BOTH SSR terminals and both indicators. R46/C38 set the window. R48 holds B low when the MCU is high-Z. R47 is the fail-safe pull-up. SJ2 = bring-up defeat, REMOVE for service.

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SSR terminals J4/J9: pin1 = SSR.EN (board +5V, watchdog-gated), pin2 = the switched low side, Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance. NOT opto-isolated – rev B tried that and reverted it: an opto only isolates if the SSR loop is powered off-board, and this board powers it.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- – both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required: two grounded-junction probes in one kIn would loop through it.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- - both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required: two grounded-junction probes in one kiln would loop through it.

Nav switch J6 is panel-mounted; inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND = dry contact each channel to GND.

Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=CS 4=RST 5=DC 6=SDI/MOSI 7=SCK

9=SD0/MISO 10=T_CLK 11=T_CS 12=T_DIN 13=T_DO 14=T_IRQ.
PIN 1 IS +5V, NOT 3V3 – the module regulates on board; do not

wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31856s (R39–R43 damping).

J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N – one CT clamp per SSR zone.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted
= do not wire to AC mains. Y1/C25/C26 load caps are an ASSUM

value (unverified C_L), see design.py comment. ADE7953 I2C address 0x38 collides with PCF8574A.